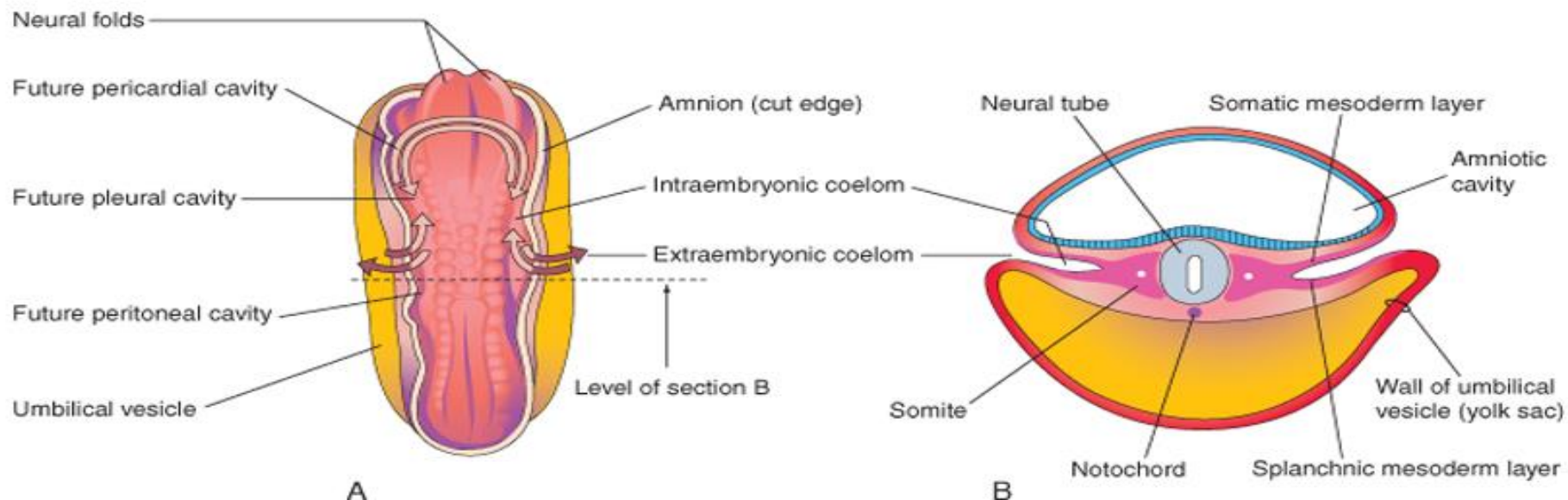


DEVELOPMENT OF BODY CAVITIES, DIAPHRAGM

DEVELOPMENT OF BODY CAVITIES

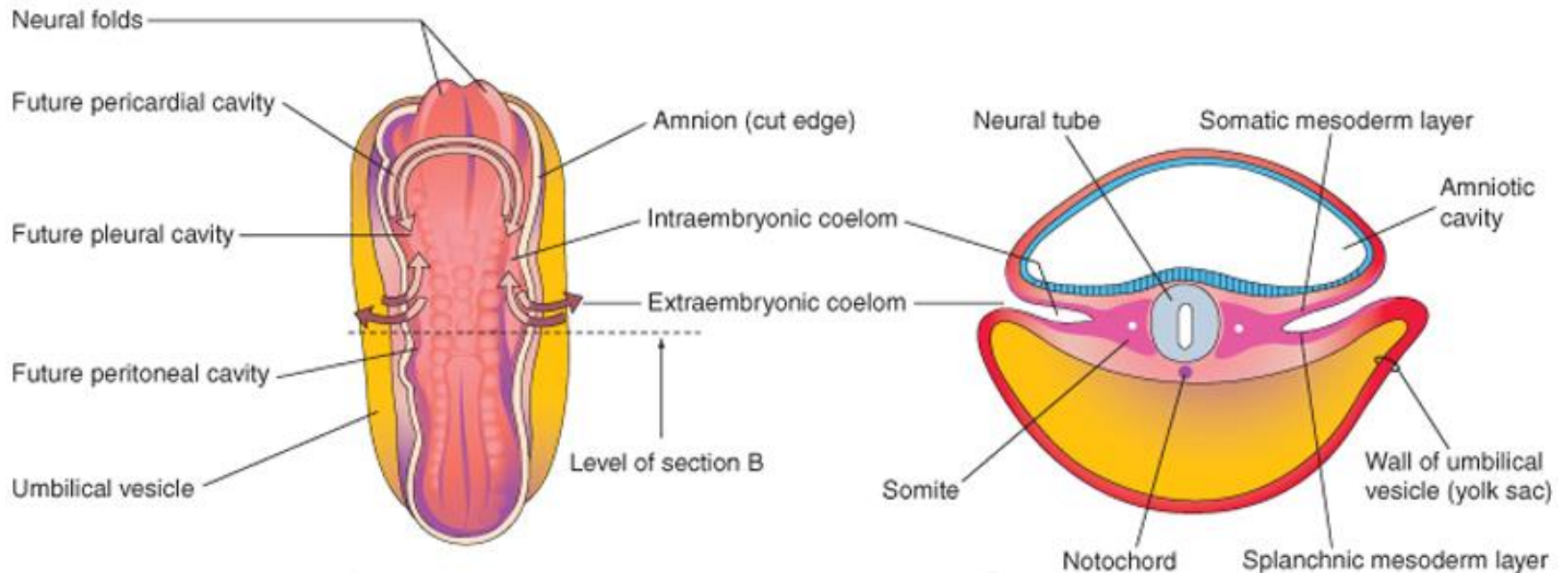
- the **intraembryonic coelom** appears as a **horseshoe-shaped** cavity early in the **4th** wk
 - The curve or bend in this cavity at the cranial end of the embryo represents the future **pericardial cavity** and its limbs (lateral extensions) indicate the future **pleural** and **peritoneal cavities**
 - The distal part of each limb of the intraembryonic coelom is continuous with the **extraembryonic coelom** at the lateral edges of the embryonic disc
 - This communication is important because most of the **midgut normally herniates** through this communication into the umbilical cord



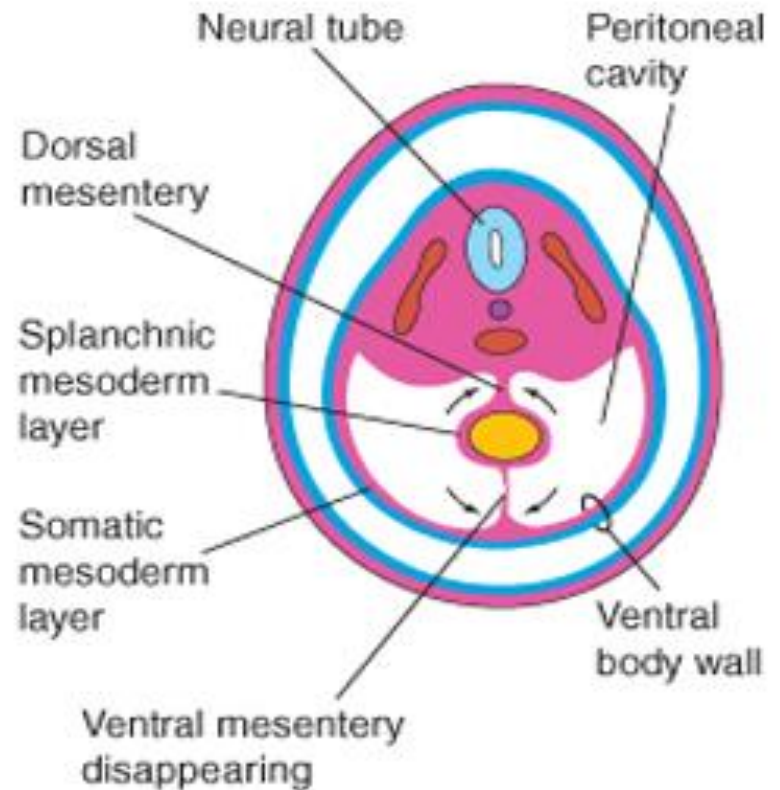
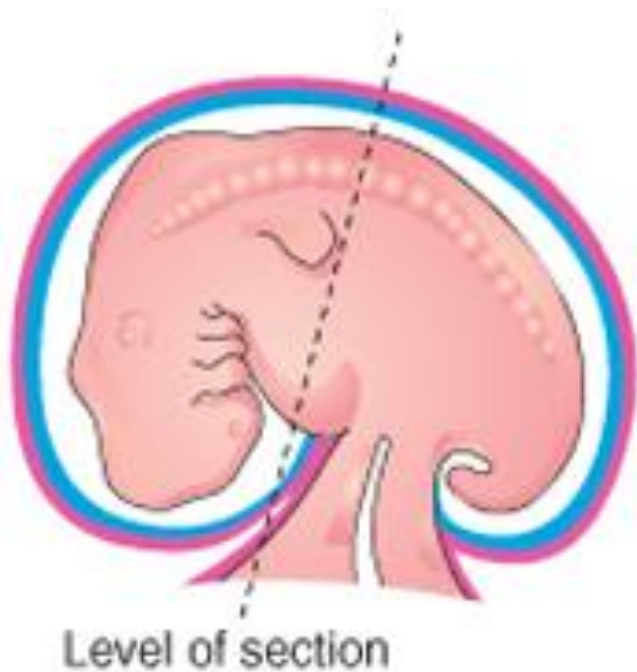
- The **intraembryonic coelom** provides room for the organs to develop and move.
- During embryonic folding in the horizontal plane, the limbs of the coelom are brought together on the ventral aspect of the embryo
- The ventral mesentery degenerates in the region of the future peritoneal cavity , resulting in a large embryonic **peritoneal cavity** extending from the heart to the pelvic region.

THE EMBRYONIC BODY CAVITY

- The **intraembryonic coelom** becomes the embryonic body cavity, which is divided into **three well-defined cavities during the 4th wk**:
 - a. A pericardial cavity
 - b. Two pericardioperitoneal canals
 - c. A peritoneal cavity

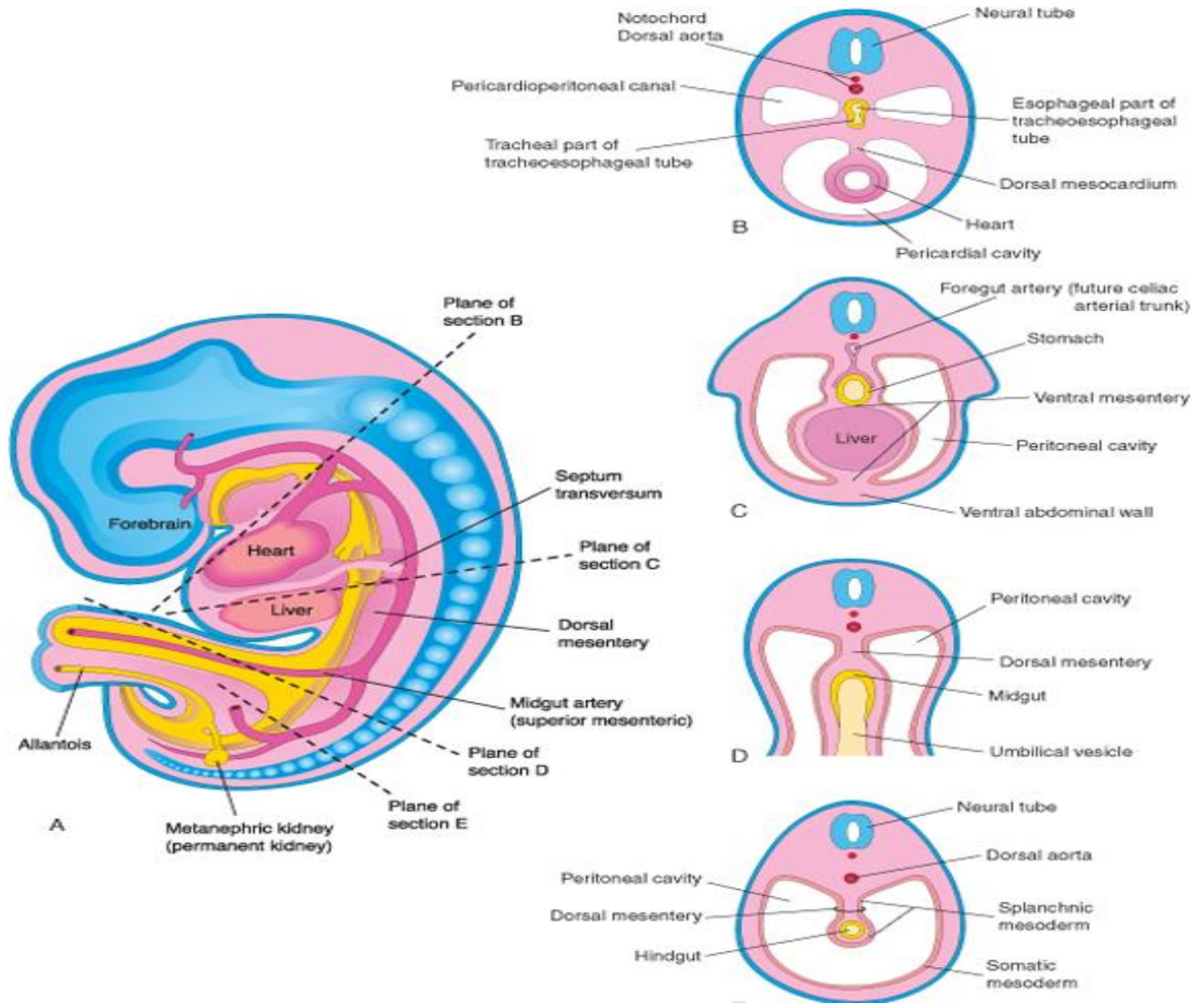


- **These body cavities have:**
 - a parietal wall lined by mesothelium (future parietal layer of peritoneum) derived from **somatic mesoderm**
 - a visceral wall covered by mesothelium (future visceral layer of peritoneum) derived from **splanchnic mesoderm**



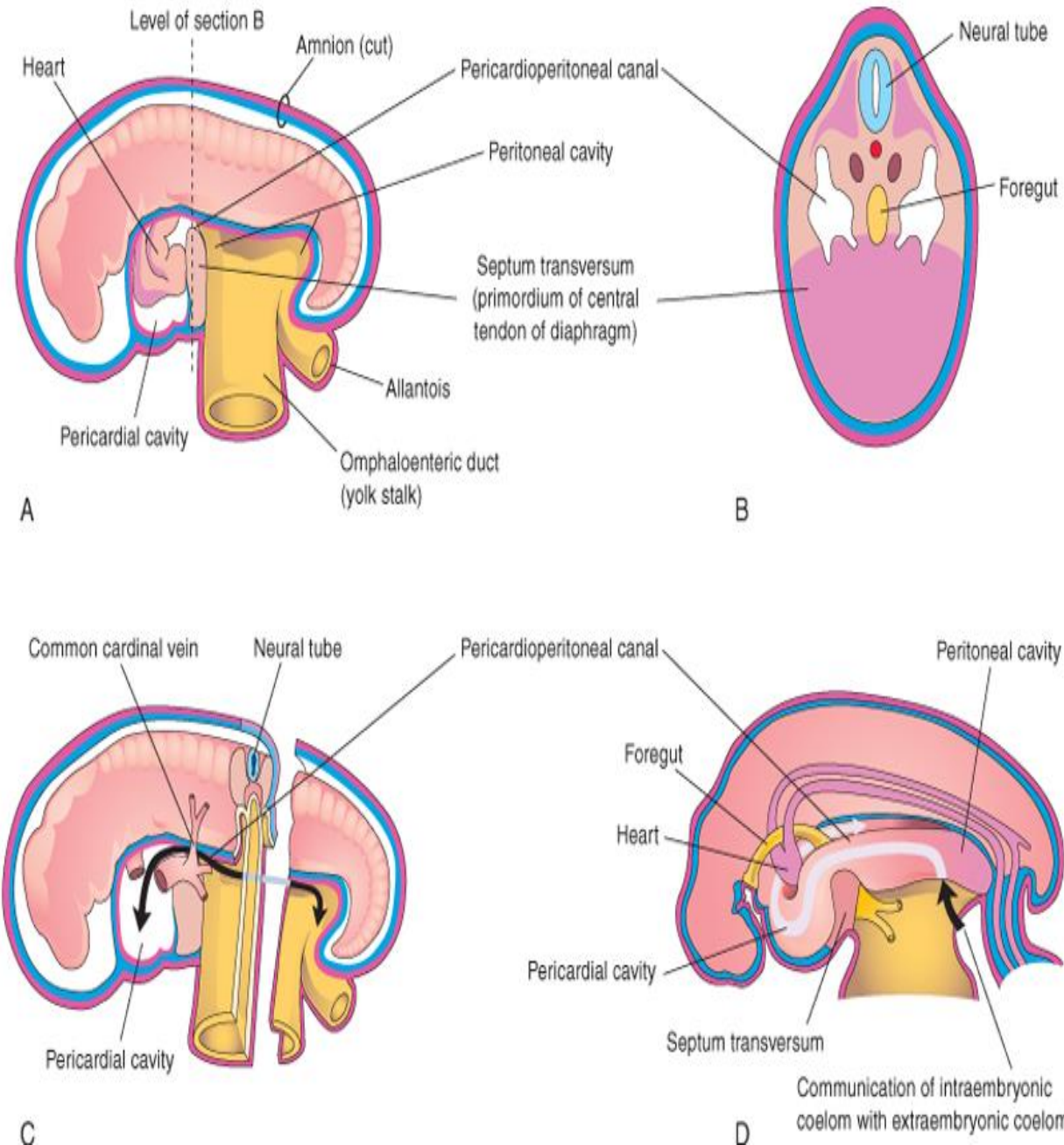
– The peritoneal cavity:

- the major part of **intraembryonic coelom** connected with the **extraembryonic coelom** at the **umbilicus loses its connection** with the extraembryonic coelom during the 10th week as the intestines return to the abdomen from the umbilical cord



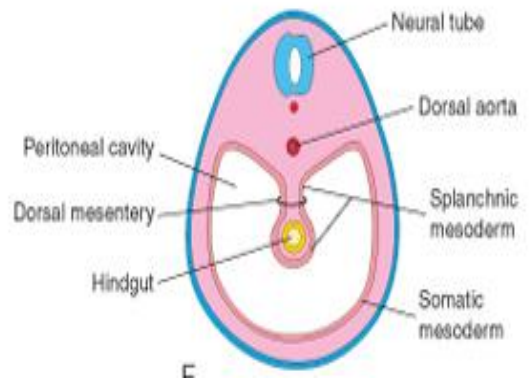
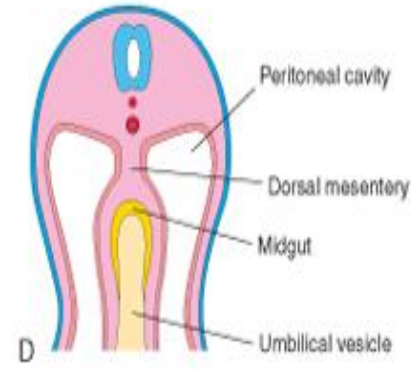
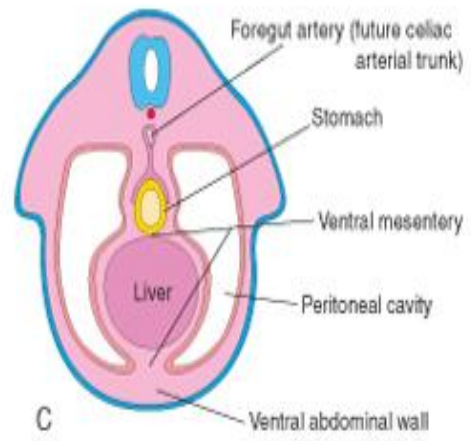
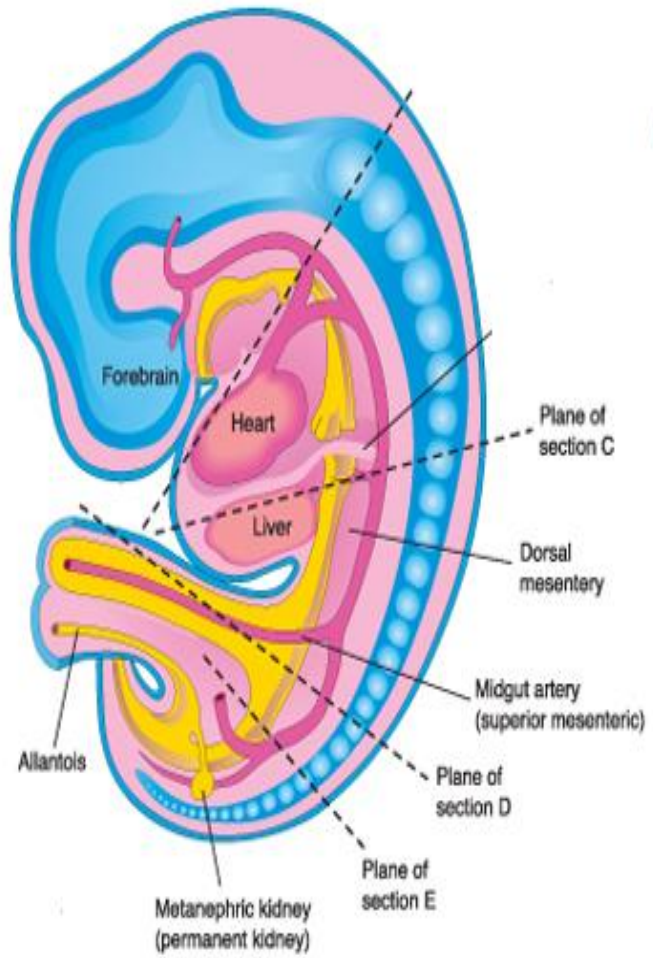
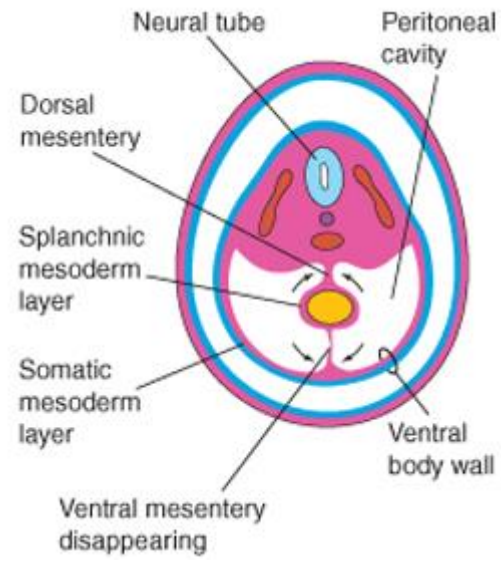
During the head fold:

- the heart and pericardial cavity are relocated **ventrocaudally, anterior to the foregut**
- the pericardial cavity opens into **pericardioperitoneal canals**, which pass dorsal to the foregut



After embryonic folding:

- the caudal part of the **foregut**, the **midgut**, and the **hindgut** are suspended in the **peritoneal cavity** from the **dorsal abdominal wall** by the **dorsal mesentery**



Mesenteries

- a double layer of peritoneum that begins as an extension of the **visceral peritoneum covering an organ**.
- connects the organ to the body wall and conveys vessels and nerves to it
- the **dorsal and ventral mesenteries** divide the **peritoneal cavity** into right and left halves
- The ventral mesentery soon disappears, except where it is attached to the caudal part of the foregut (**primordium of stomach and proximal part of duodenum**).
- The peritoneal cavity then becomes a continuous space
- The arteries supplying the primordial **gut-celiac trunk** (foregut), **SMA** (midgut), and **IMA** (hindgut)-pass between the layers of the **dorsal mesentery**

Division of the embryonic body cavity

- Each **pericardioperitoneal canal**:
 - lies lateral to the proximal part of the foregut (future esophagus) and dorsal to the **septum transversum**
 - The **septum transversum** is the primordium of the central tendon of the diaphragm.
 - **Partitions form in each pericardioperitoneal canal**:
 - separate the **pericardial cavity** from the **pleural cavities**
 - the **pleural cavities** from the **peritoneal cavity**.

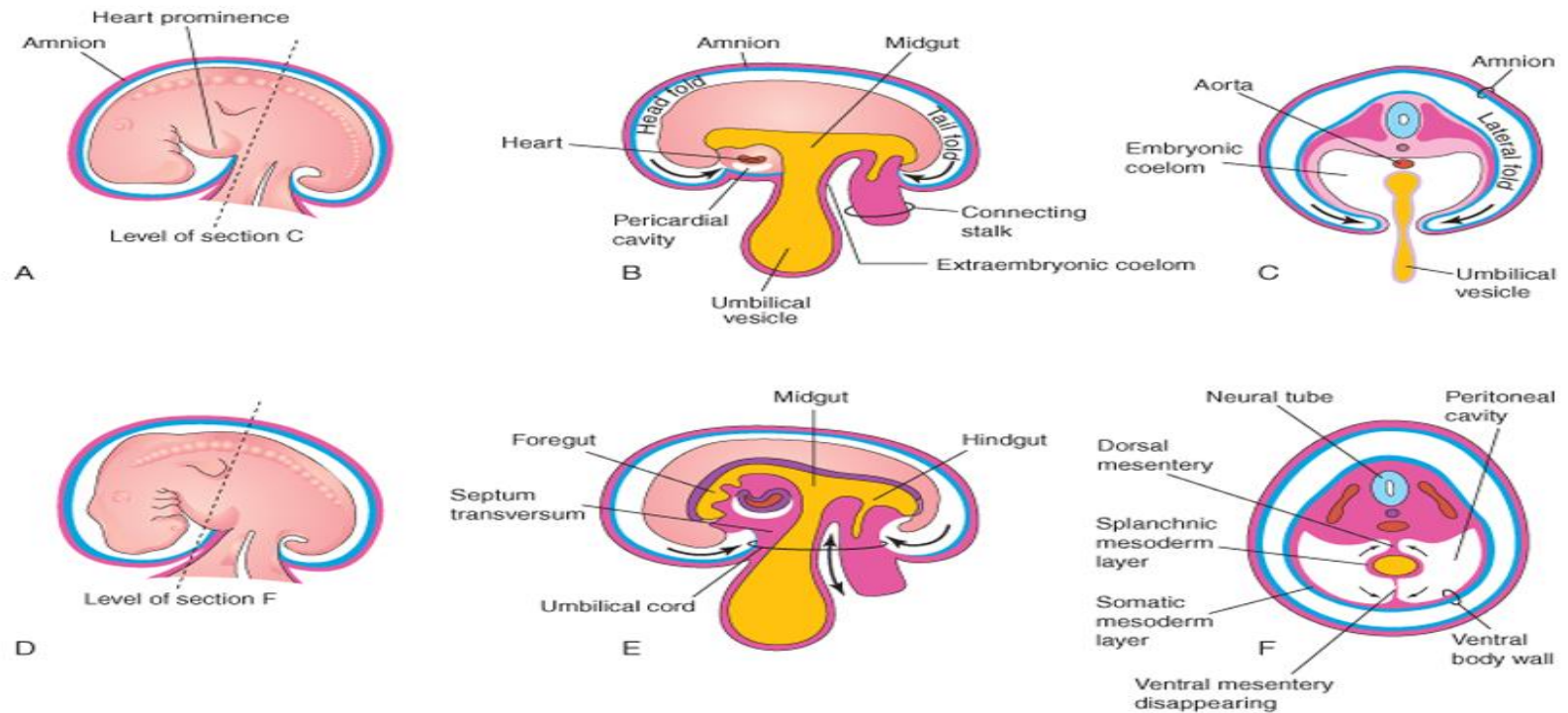


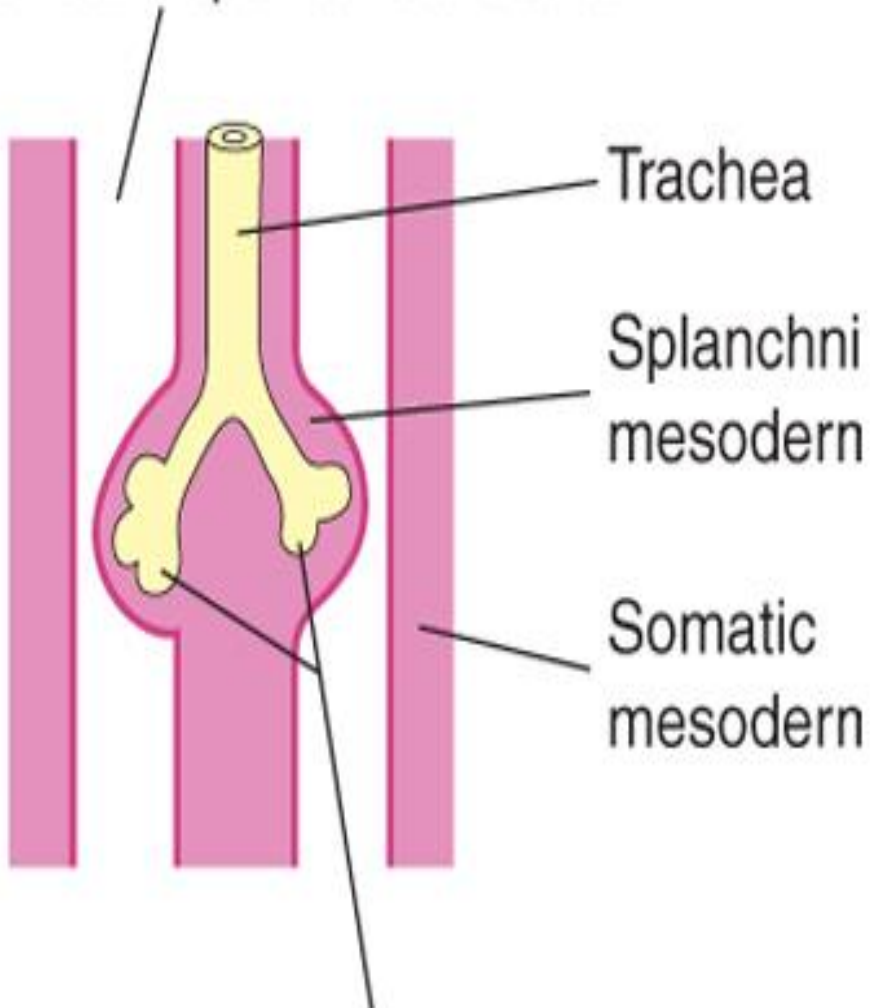
Figure 8-2 Illustrations of embryonic folding and its effects on the intraembryonic coelom and other structures. **A**, Lateral view of an embryo (**approximately 26 days**). **B**, Schematic sagittal section of this embryo showing the **head and tail folds**. **C**, Transverse section at the level shown in **A**, indicating how fusion of **the lateral folds** gives the embryo a **cylindrical form**. **D**, Lateral view of an embryo (**approximately 28 days**). **E**, Schematic sagittal section of this embryo showing the reduced communication between the intraembryonic and extraembryonic coeloms (**double-headed arrow**). **F**, Transverse section as indicated in **D**, illustrating formation of the ventral body wall and disappearance of the ventral mesentery. The *arrows* indicate the junction of the somatic and splanchnic layers of mesoderm. The somatic mesoderm will become the parietal peritoneum lining the abdominal wall, and the splanchnic mesoderm will become the visceral peritoneum covering the organs (e.g., the stomach).

- As the **bronchial buds** (primordia of bronchi and lungs) grow into the **pericardioperitoneal canals**:
 - a pair of membranous ridges is produced in the **lateral wall of each canal**:
 - The cranial ridges - *pleuropericardial folds* - are located superior to the developing lungs.
 - The caudal ridges - *pleuroperitoneal folds* - are located inferior to the lungs

- **pleuropericardial folds**

- **is** flaps of body wall at the level of the primitive pericardial cavity which grow medially
- and fuse with the ventral surface of the foregut mesoderm
- This subdivides the primitive pericardial cavity into the definitive **pericardial** and **two pleural cavities**.

Pericardioperitoneal canal



Trachea

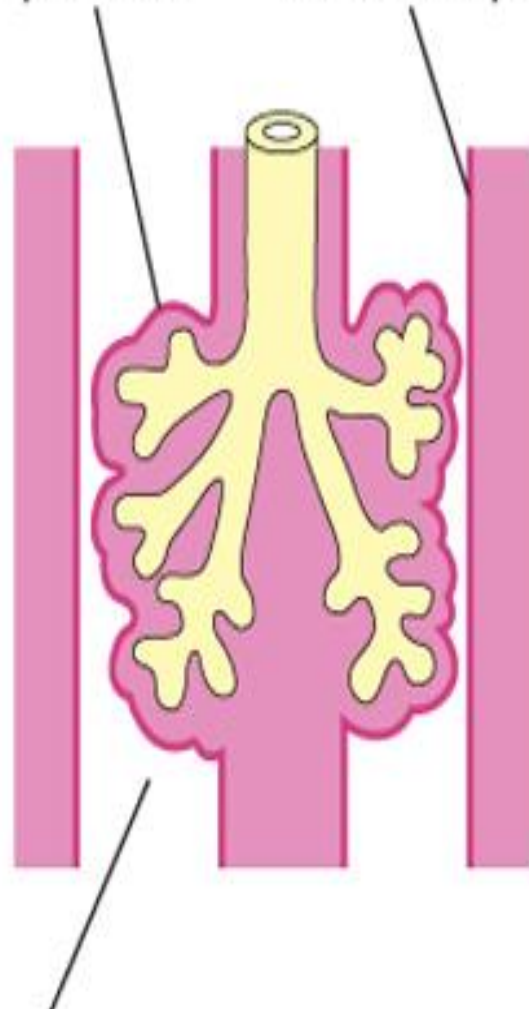
Splanchni
mesoderm

Somatic
mesoderm

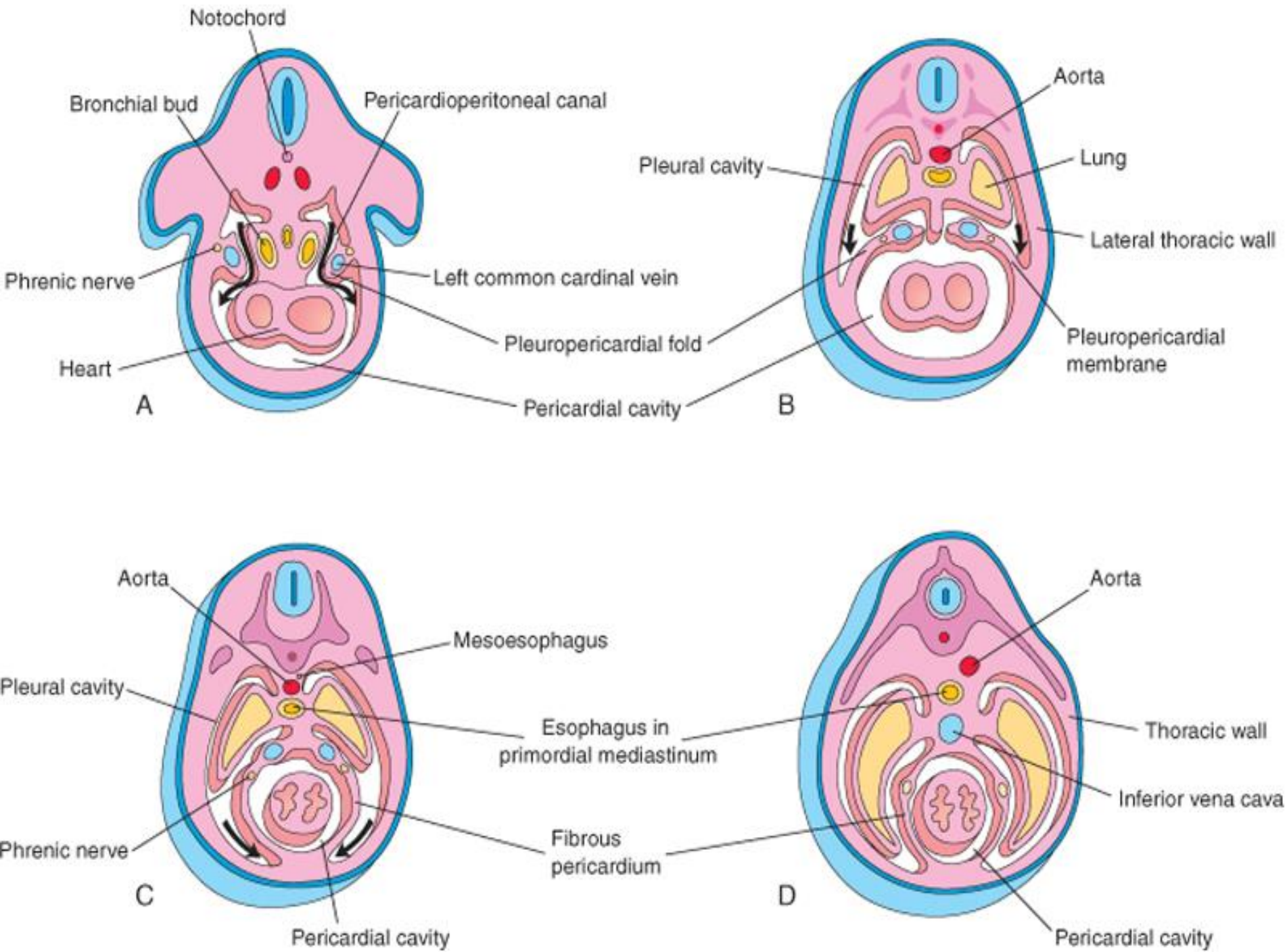
Secondary bronchial buds

Visceral pleura

Parietal pleura



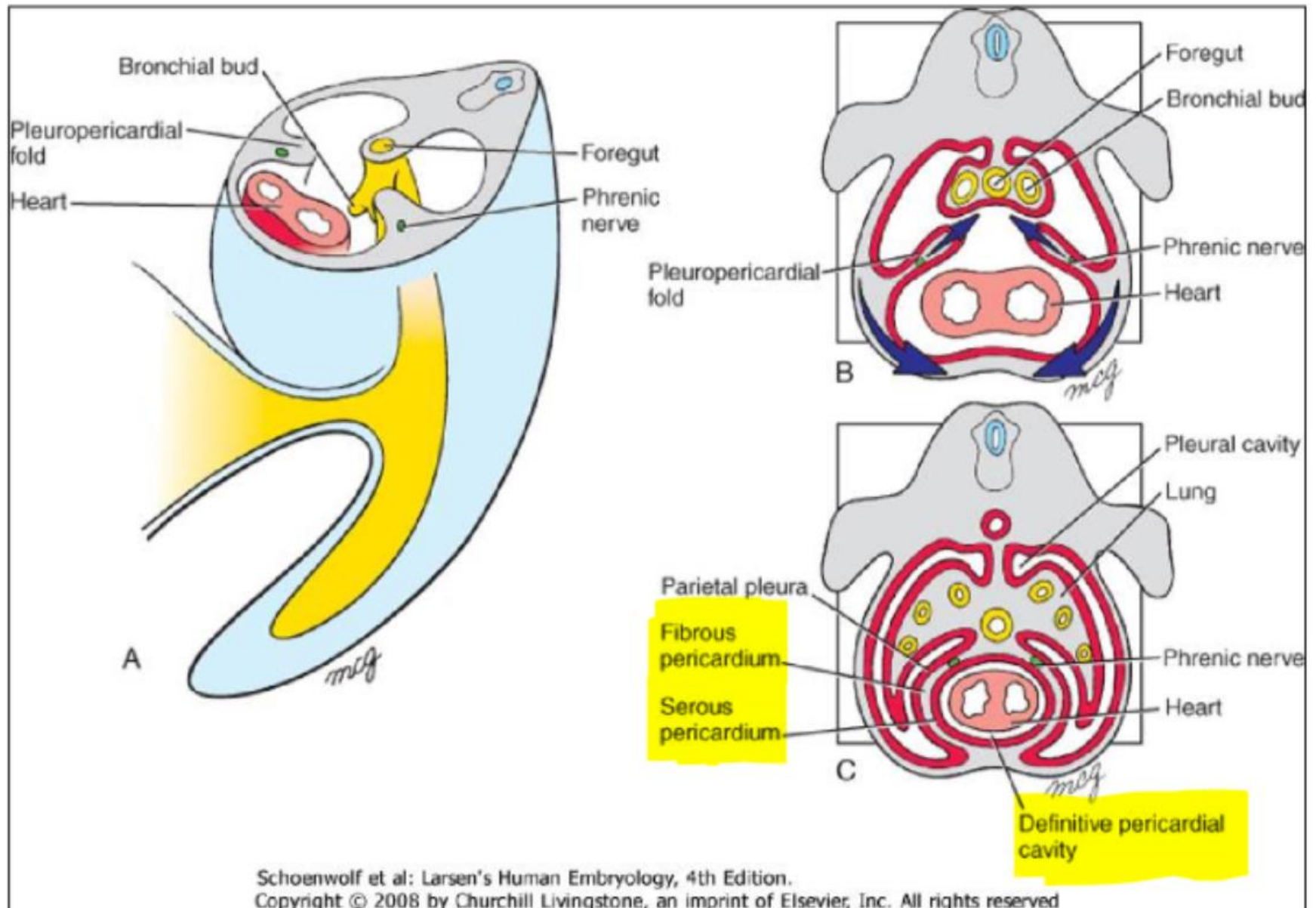
Pleural cavity

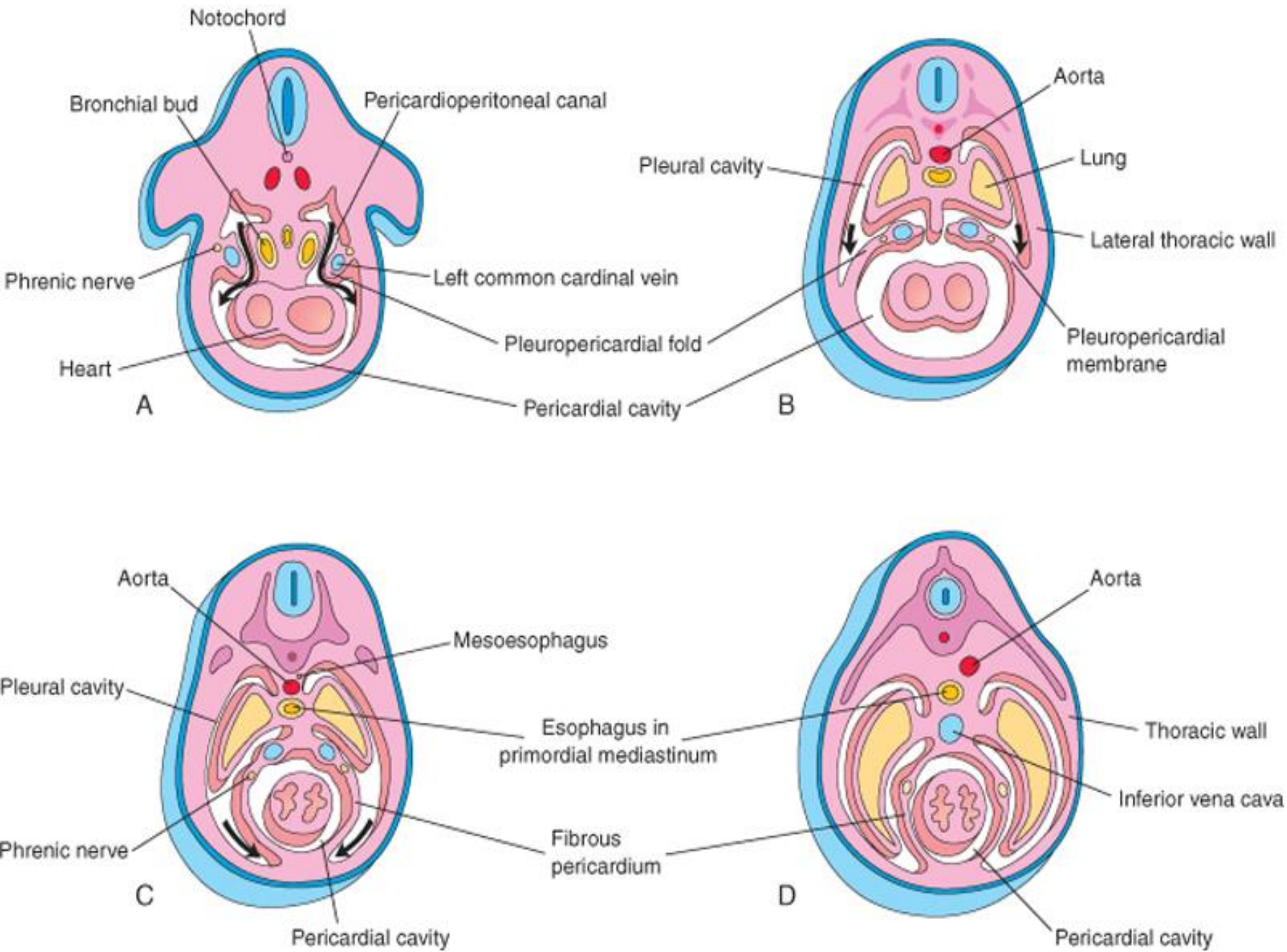


Pleuropericardial Membranes

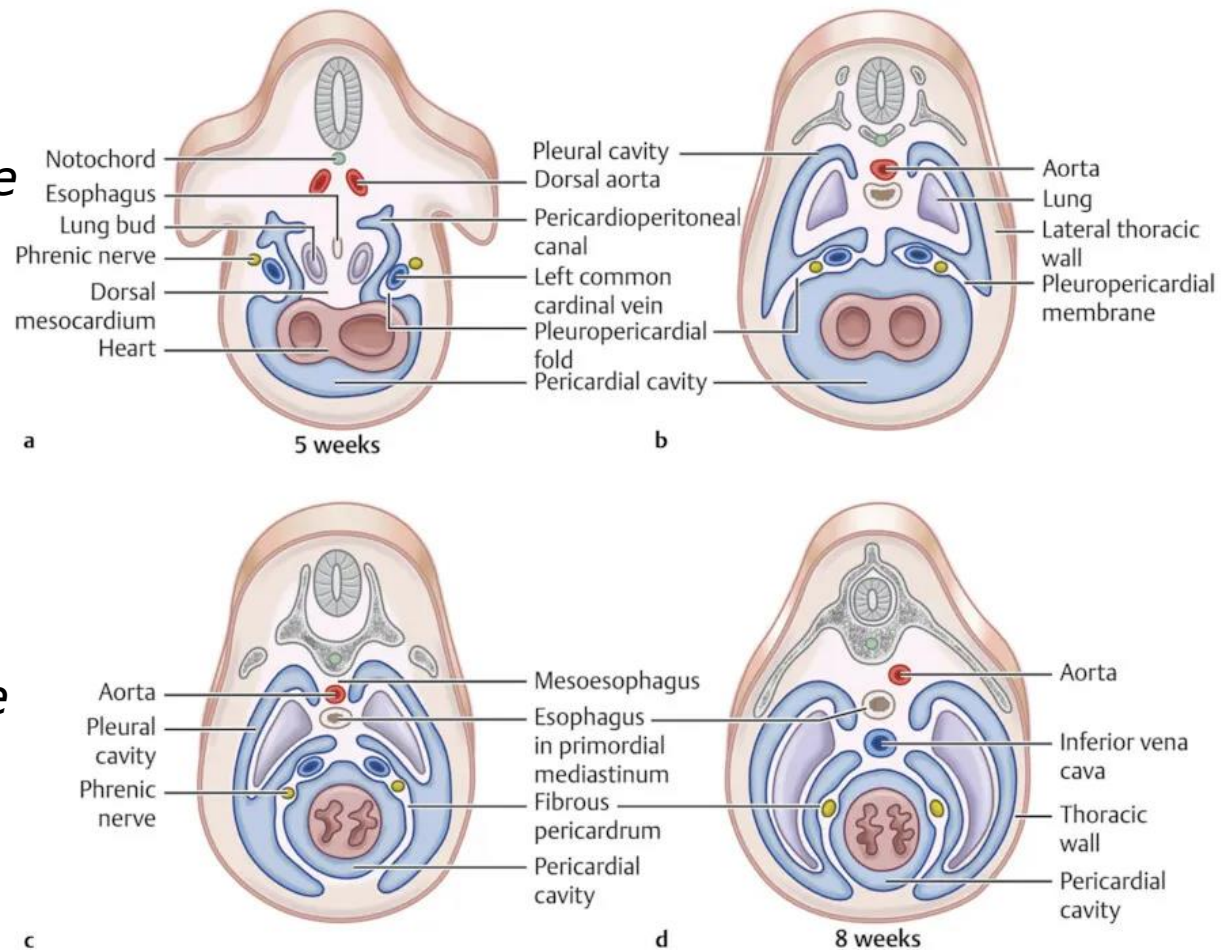
- As the **pleuropericardial folds enlarge**, they form partitions that separate the **pericardial cavity** from the **pleural cavities**
- These partitions-**pleuropericardial membranes** contain the common cardinal veins, which drain the venous system into the sinus venosus of the heart
- Initially the **bronchial buds** are small relative to the heart and pericardial cavity
- They soon grow laterally from the caudal end of the trachea into the pericardioperitoneal canals (future pleural canals).
- As the **primordial pleural cavities expand ventrally around the heart**, they extend into the body wall, splitting the mesenchyme into:
 - **An outer layer** that becomes the **thoracic wall**
 - **An inner layer** (pleuropericardial membrane) that becomes **the fibrous pericardium**

Subdivision of the Pericardial Cavity





(a-d) At 5 weeks, the pleuropericardial folds appear and by 6 weeks reach midline and later fuse with the splanchnic mesoderm (primitive mediastinum) at the lung hila to separate the pleural spaces from the pericardial cavity. The dorsal mesocardium connects the heart to the splanchnic mesoderm that overlies the embryonic pharynx/esophagus. This mesocardium is the site through which vessels or additional tissue can dorsally enter the heart. The lung buds develop from the foregut and grow into the pericardioperitoneal canals to form the pleural cavities. The cardinal veins are located in these folds and later transform into the sinus venosus of the heart. Along with growth of the pleural cavities, the mediastinal mesenchyma will be pushed against the pleural cavity to form the fibrous pericardium with the phrenic neurovascular bundle embedded in it.

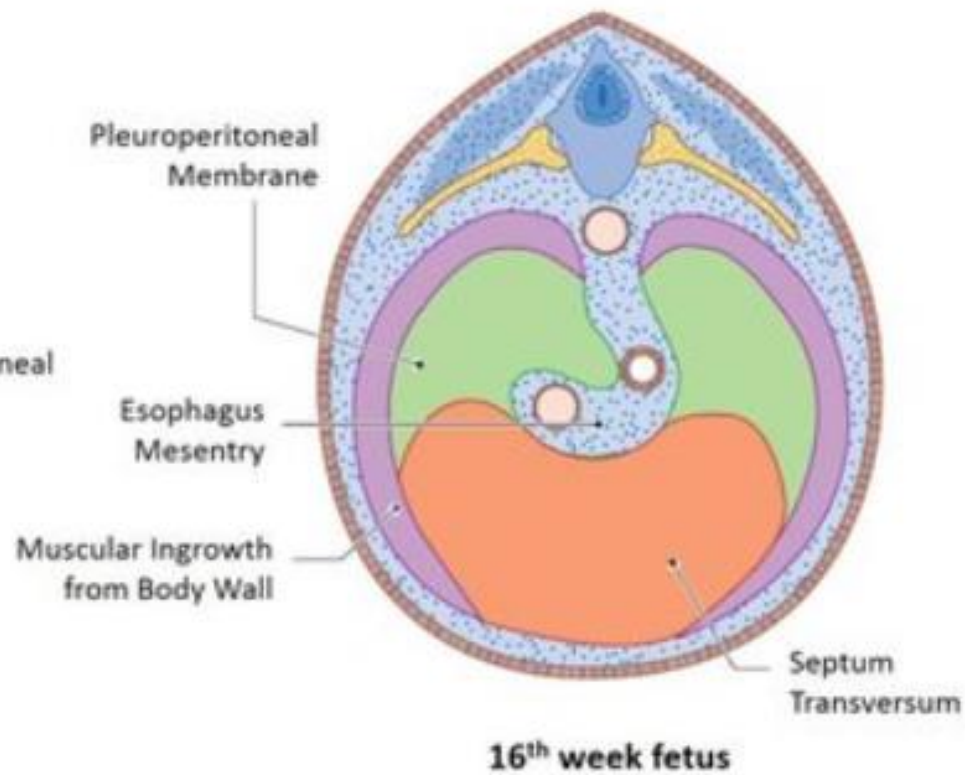
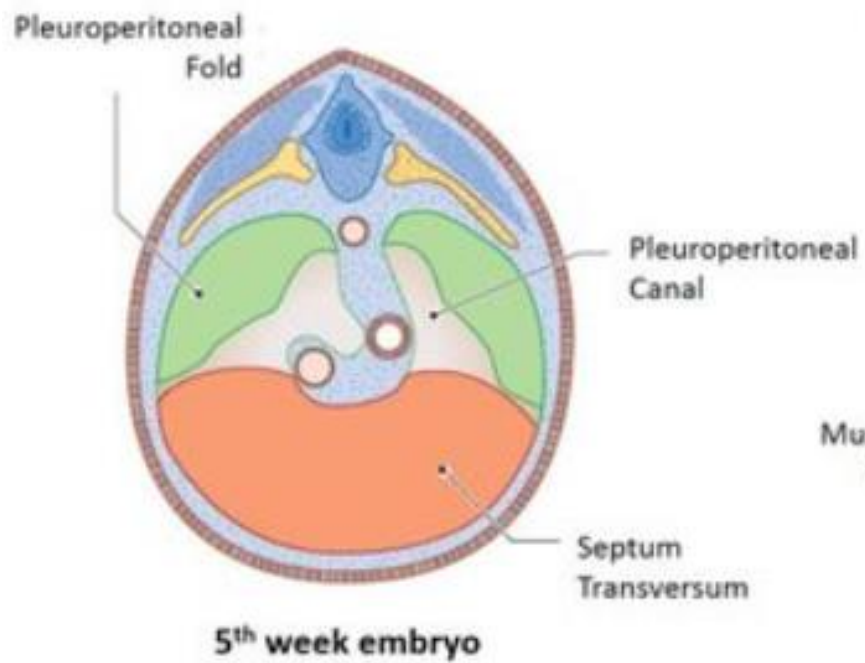


- **The pleuropericardial membranes:**
 - project into the cranial ends of the pericardioperitoneal canals
 - **With subsequent growth of:**
 - the common cardinal veins,
 - positional displacement of the heart,
 - and expansion of the pleural cavities, the pleuropericardial membranes become **mesentery-like folds extending from the lateral thoracic wall.**
 - **By the 7th week,**
 - the pleuropericardial membranes fuse with the mesenchyme ventral to the esophagus, separating the **pericardial cavity from the pleural cavities**

- **This primordial mediastinum:**
 - consists of a mass of mesenchyme that extends from the sternum to the vertebral column, separating the developing lungs
- **The right pleuropericardial opening** closes slightly earlier than the left one and produces a larger pleuropericardial membrane

Pleuroperitoneal Membranes

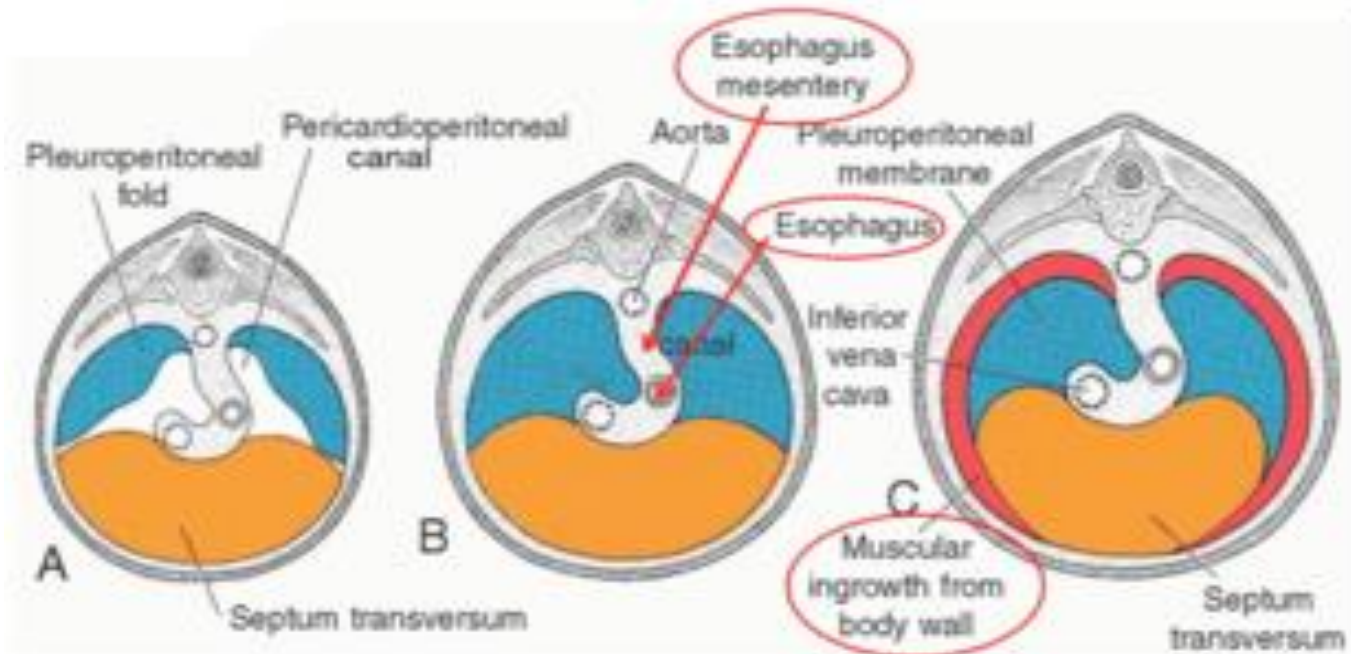
- As the **pleuroperitoneal folds** enlarge, they project into the **pericardioperitoneal canals**
- the folds become membranous, forming the **pleuroperitoneal membranes**
- these membranes separate the **pleural cavities** from the **peritoneal cavity**.
- **The pleuroperitoneal membranes:**
 - are produced as the developing lungs and pleural cavities expand and invade the body wall.
 - attached dorsolaterally to the abdominal wall
 - and initially their crescentic free edges project into the caudal ends of the **pericardioperitoneal canals**



- During the 6th week, the pleuroperitoneal membranes extend ventromedially until their free edges fuse with the dorsal mesentery of the esophagus and septum transversum
- This separates the pleural cavities from the peritoneal cavity.
- Closure of the pleuroperitoneal openings is assisted by the migration of myoblasts (primordial muscle cells) into the pleuroperitoneal membranes
- The pleuroperitoneal opening on the right side closes slightly before the left one.
 - The reason for this is uncertain, but it may be related to the relatively large size of the right lobe of the liver at this stage of development.

DEVELOPMENT OF THE DIAPHRAGM

- The diaphragm develops from four embryonic components:
 1. Septum transversum
 2. Pleuroperitoneal membranes
 3. Dorsal mesentery of esophagus
 4. Muscular ingrowth from lateral body walls



1. Septum Transversum:

- composed of mesodermal tissue
- is the primordium of the **central tendon of the diaphragm**
- grows **dorsally from the ventrolateral body wall** and forms a semicircular shelf, which separates the heart from the liver
- is first identifiable at **the end of the 3rd week** as a mass of mesodermal tissue cranial to the pericardial cavity
- After the head folds ventrally **during the 4th week**, the septum transversum forms a thick incomplete partition between **the pericardial and abdominal cavities**

- **The septum transversum:**

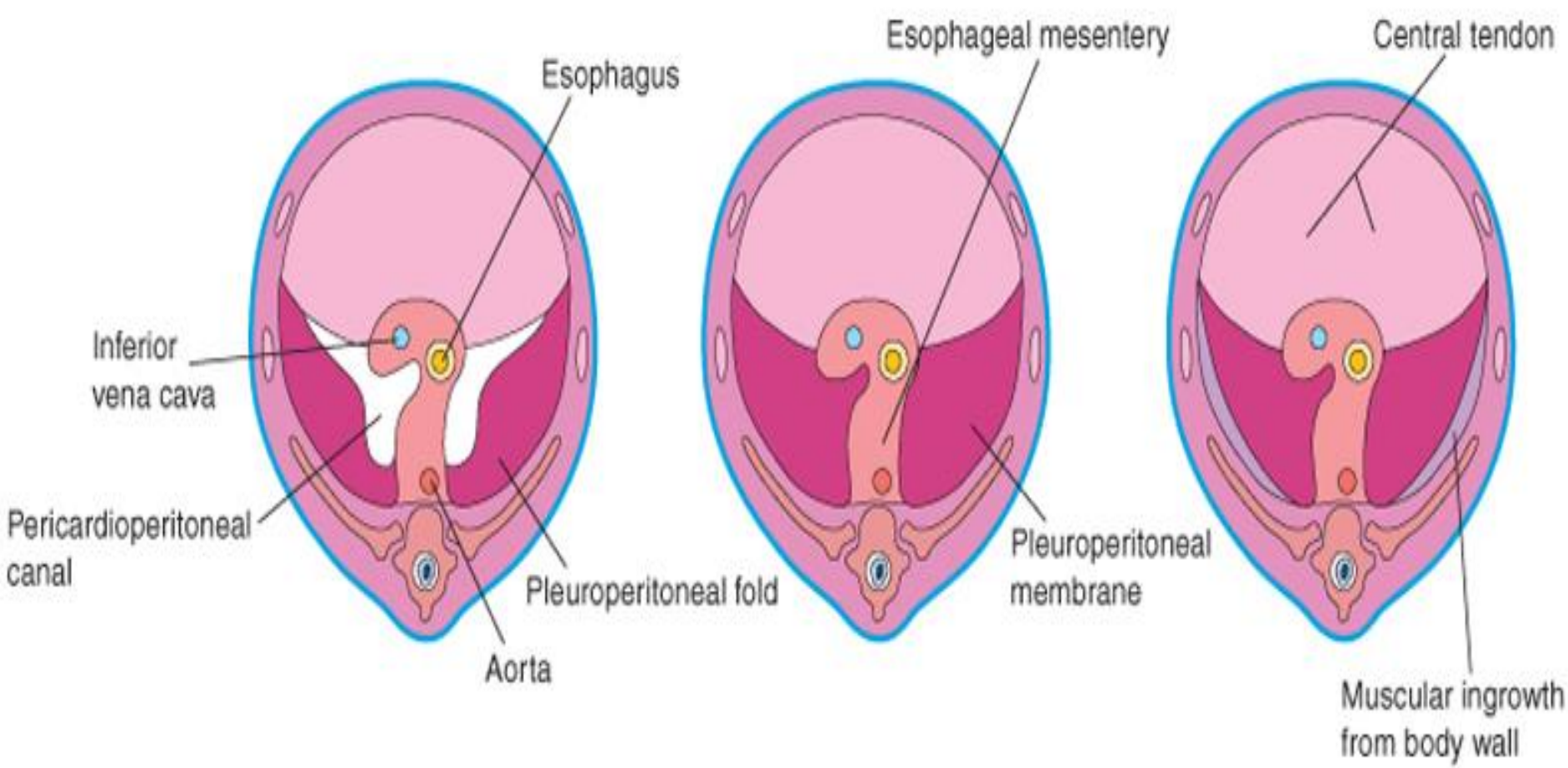
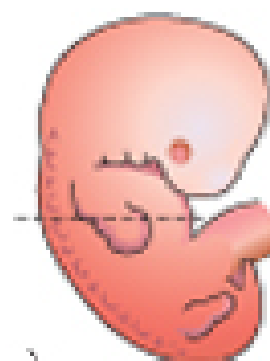
- does not separate the thoracic and abdominal cavities completely.
- during its early development, a large part of the liver is embedded in the septum transversum.
- there are large openings, the **pericardioperitoneal canals**, along the sides of the esophagus
- expands and fuses with the dorsal mesentery of the esophagus and the pleuroperitoneal membranes

Pleuroperitoneal Membranes:

- Fuse with the dorsal mesentery of the esophagus and the septum transversum
- This completes the partition between the thoracic and abdominal cavities and forms the **primordial diaphragm**.
- Although the **pleuroperitoneal membranes** form large portions of the early fetal diaphragm, they represent relatively small portions of the newborn's diaphragm

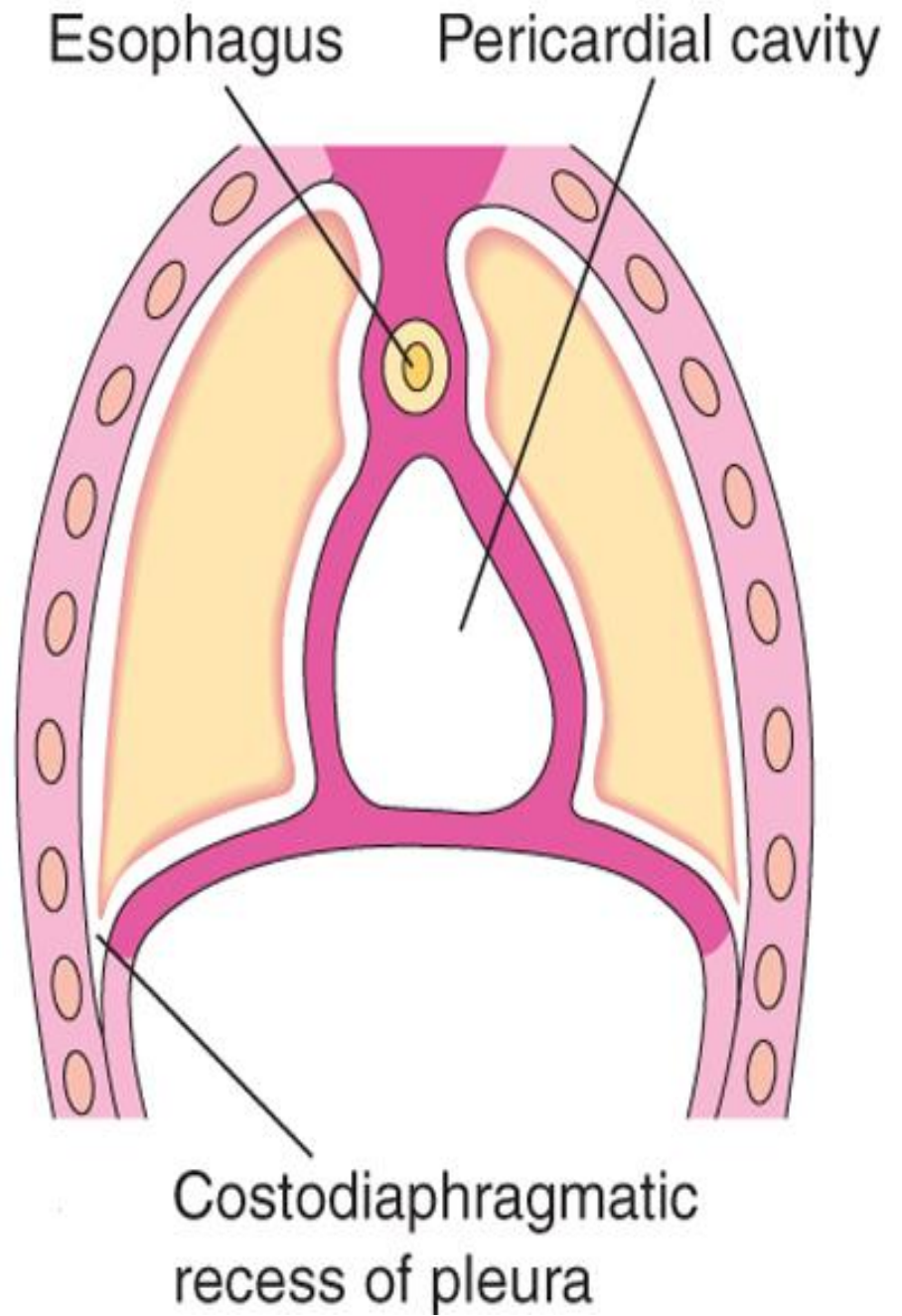
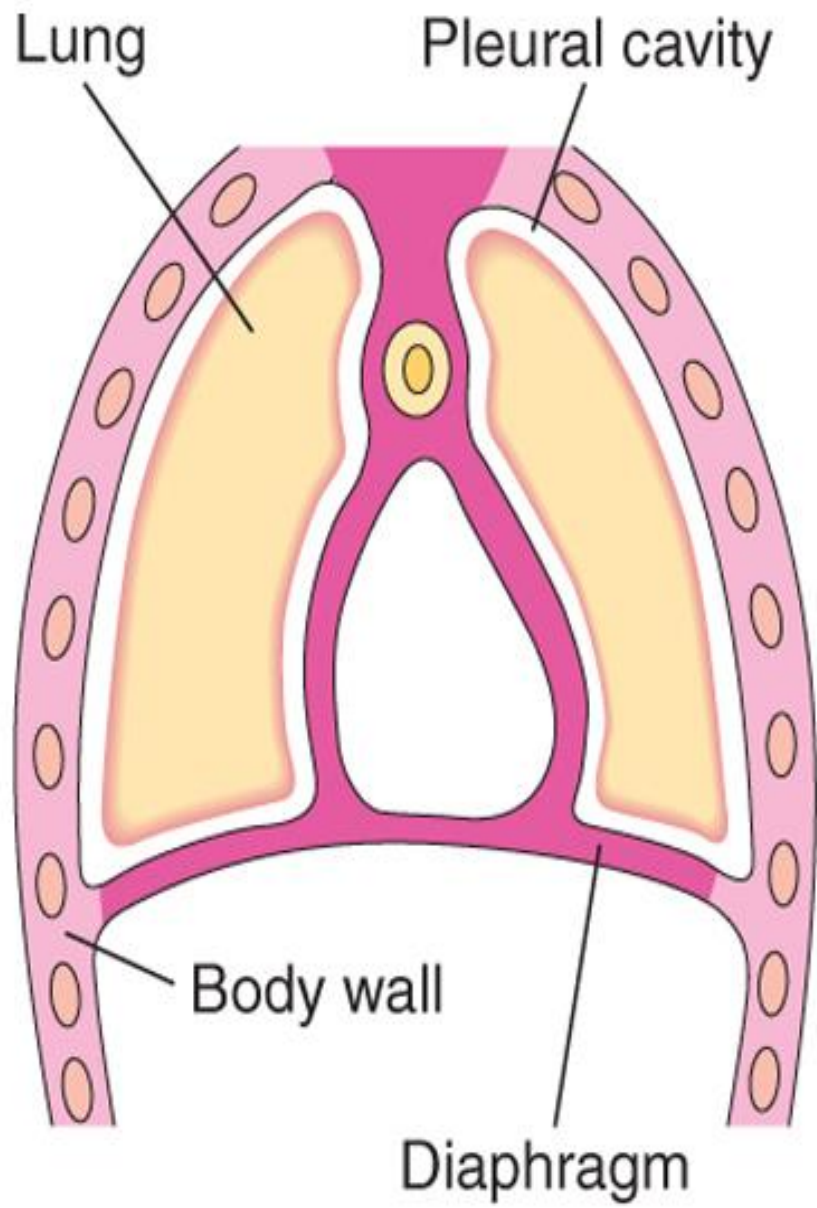
Dorsal mesentery of the esophagus:

- This mesentery constitutes the median portion of the diaphragm.
- The **crura of the diaphragm**, a leglike pair of diverging muscle bundles that cross in the median plane anterior to the aorta, develop from **myoblasts** that grow into the **dorsal mesentery of the esophagus**.



Muscular in growth from lateral body walls:

- During the 9th to 12th weeks, the lungs and pleural cavities enlarge, "burrowing" into the lateral body walls
- As result the body-wall tissue is split into two layers:
 - **An external layer** that becomes part of the definitive abdominal wall
 - **An internal layer** that contributes to peripheral parts of the diaphragm, external to the parts derived from the pleuroperitoneal membranes
- Further extension of the developing pleural cavities into the lateral body walls forms the right and left **costodiaphragmatic recesses**, establishing the characteristic dome-shaped configuration of the diaphragm.
- After birth, the costodiaphragmatic recesses become alternately smaller and larger as the lungs move in and out of them during inspiration and expiration.



Positional changes and innervation of the diaphragm

- During the 4th week, the septum transversum- lies opposite the 3rd to 5th cervical somites
- During the 5th week, **myoblasts** from these somites migrate into the developing diaphragm, bringing their nerve fibers with them
- the **phrenic nerves** that supply motor innervation to the diaphragm arise from the ventral primary rami of the **C3, C4, and C5 spinal nerves**.
- The three twigs on each side join together to form a phrenic nerve.
- The phrenic nerves also supply sensory fibers to the superior and inferior surfaces of the right and left domes of the diaphragm

- By the **6th week**, the developing diaphragm is at the level of **the thoracic somites**
- By the beginning of the **8th week**, the dorsal part of the diaphragm lies at the level of the **first lumbar vertebra**
- The phrenic nerves in the embryo enter the diaphragm by passing through the **pleuropericardial membranes**.
 - This explains why the phrenic nerves subsequently lie on the fibrous pericardium, the adult derivative of the **pleuropericardial membranes**
 - As the four parts of the diaphragm fuse, mesenchyme in the **septum transversum** extends into the other three parts.



Septum transversum



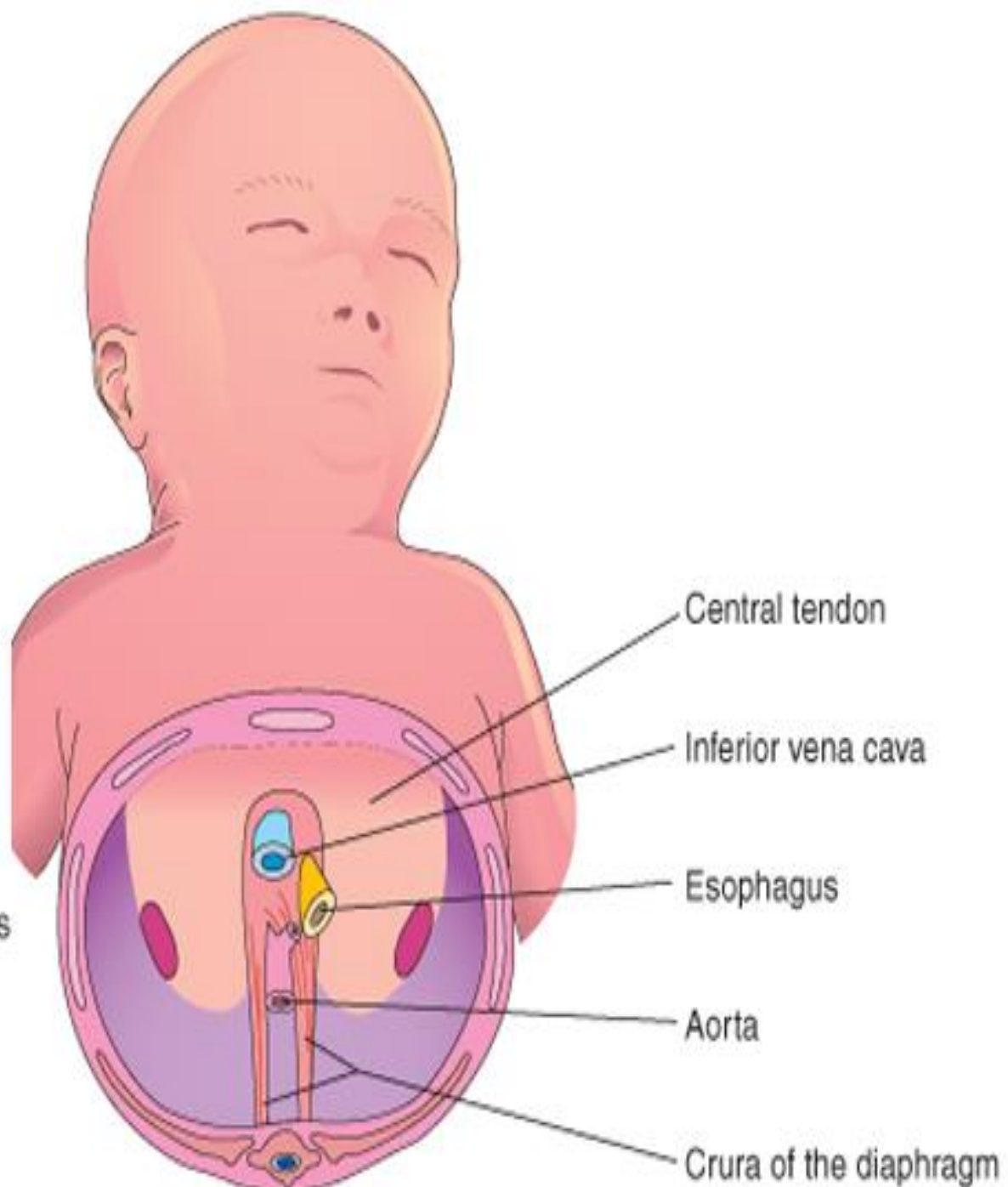
Mesentery of esophagus



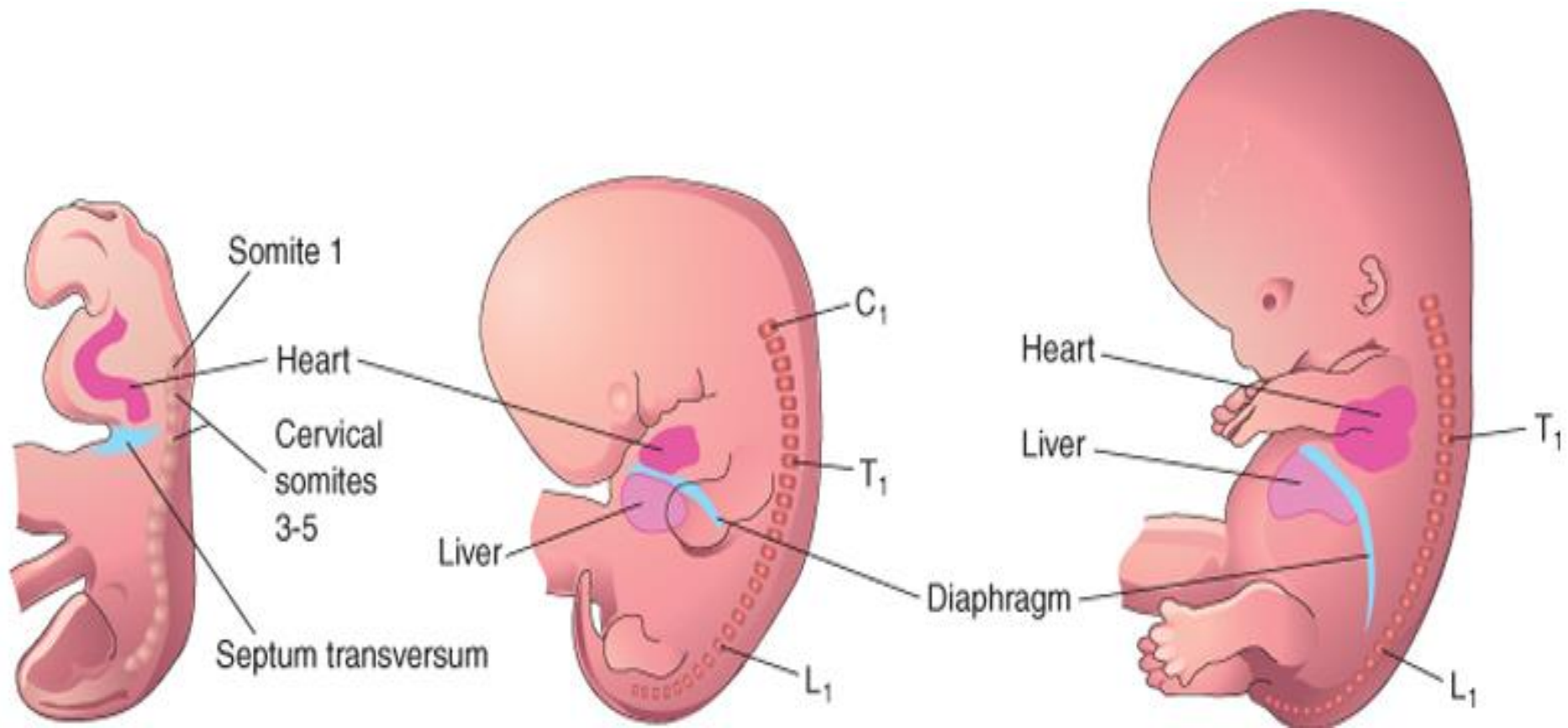
Pleuroperitoneal folds and membranes



Muscular ingrowth from body wall



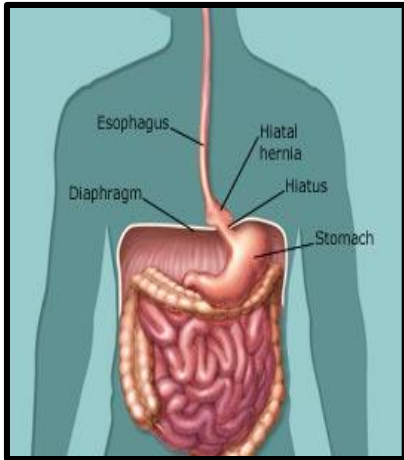
- It forms myoblasts that differentiate into the skeletal muscle of the diaphragm.
- The costal border receives sensory fibers from the **lower intercostal nerves** because of the origin of the peripheral part of the diaphragm from the lateral body walls



Congenital Anomalies

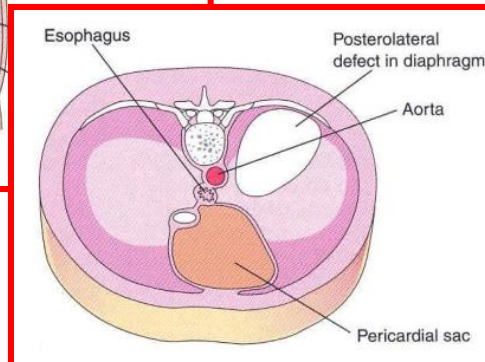
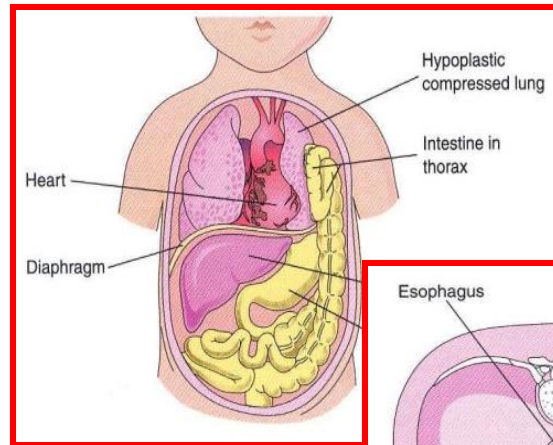
Congenital hiatal hernia:

because of large esophageal hiatus



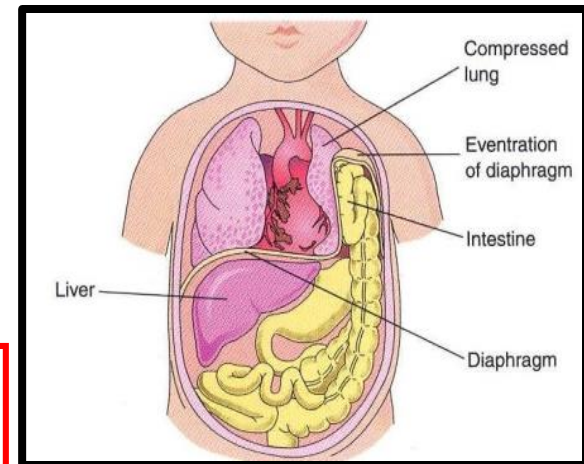
Congenital diaphragmatic hernia:

Commonly through a posterolateral defect in diaphragm. Mostly on left side. Left lung shows hypoplasia



Eventration of diaphragm:

because of defective musculature



SUMMARY OF THE DEVELOPMENT OF THE BODY CAVITIES

- The intraembryonic coelom begins to develop near the end of the third week. By the fourth week, it appears as a horseshoe-shaped cavity in the cardiogenic and lateral mesoderm. The curve of the cavity represents the future pericardial cavity and its lateral extensions represent the future pleural and peritoneal cavities.
- During folding of the embryonic disc in the fourth week, lateral parts of the intraembryonic coelom move together on the ventral aspect of the embryo. When the caudal part of the ventral mesentery disappears, the right and left parts of the intraembryonic coelom merge to form the peritoneal cavity.
- As peritoneal parts of the intraembryonic coelom come together, the splanchnic layer of mesoderm encloses the primordial gut and suspends it from the dorsal body wall by a double-layered peritoneal membrane, the dorsal mesentery.
- The parietal layer of mesoderm lining the peritoneal, pleural, and pericardial cavities becomes the parietal peritoneum, parietal pleura, and serous pericardium, respectively.
- Until the seventh week, the embryonic pericardial cavity communicates with the peritoneal cavity through paired pericardioperitoneal canals. During the fifth and sixth weeks, folds (later membranes) form near the cranial and caudal ends of these canals.
- Fusion of the cranial pleuropericardial membranes with mesoderm ventral to the esophagus separates the pericardial cavity from the pleural cavities.
- Fusion of the caudal pleuroperitoneal membranes during formation of the diaphragm separates the pleural cavities from the peritoneal cavity.

